

OVERVIEW

- SPARTA (Soil Properties Assessment, Resistance, and Thermal Analysis) is a multi-instrument regolith characterization system developed at NASA Jet Propulsion Laboratory (JPL).
- The system integrates a cone penetrometer (CPT), vane shear tool (VST), dielectric sensor (DSP), and thermal control probe (TCP) to measure mechanical, thermal, and electrical properties of planetary soils.
- SPARTA is designed for deployment across diverse planetary environments, including the Moon, Mars, Venus, asteroids, and comets.
- The platform supports in situ subsurface investigations critical to landing safety, mobility analysis, excavation, and future in-situ resource utilization (ISRU) missions.

PURPOSE

- Reliable planetary surface operations require accurate, in-situ knowledge of regolith mechanical and thermal properties, which are currently limited and uncertain.
- SPARTA was developed to address this gap by providing a robust, multi-instrument system capable of characterizing planetary soils across diverse environments.
- The system supports mission-critical decisions related to landing safety, mobility, excavation, and ISRU, reducing risk for future robotic and crewed exploration.

CONTRIBUTION

- Designed and modeled stabilizing rails in SolidWorks, produced 3D-printed prototypes, and prepared designs for future machining to reduce lateral drift during penetration and shear tests.
- Improved sensor accuracy and repeatability through automated zero/tare routines, noise filtering, and refined numerical processing; developed Python tools to parse and overlay large CSV datasets across multiple test runs.
- Contributed to vibration-enabled regolith container designs, including motor mounts, O-ring sealing, and clearance geometry for automated soil reconditioning.

CREDIT

Robert C. Anderson (Mentor)

SPARTA PROBE AND TEST BENCH



Figure 1. SPARTA Testbench on field

- Figure 1: The SPARTA testbench is a vertically integrated actuation and sensing assembly that applies controlled axial and torsional loads during subsurface interaction tests. A linear actuator drives penetration force measurement via a load cell, while a planetary gear reduction stage provides torque. The gearbox output is instrumented with a torque sensor, with the SPARTA probe mounted at the tip for simultaneous force and torque measurements.
- Figure 2: The SPARTA multi-instrument probe integrates a Cone Penetrometer (CPT), Vane Shear Tool (VST), Dielectric Sensor Probe (DSP), and Thermal Control Probe (TCP) in a compact assembly, enabling simultaneous mechanical, electrical, and thermal characterization of planetary regolith in a single deployment.

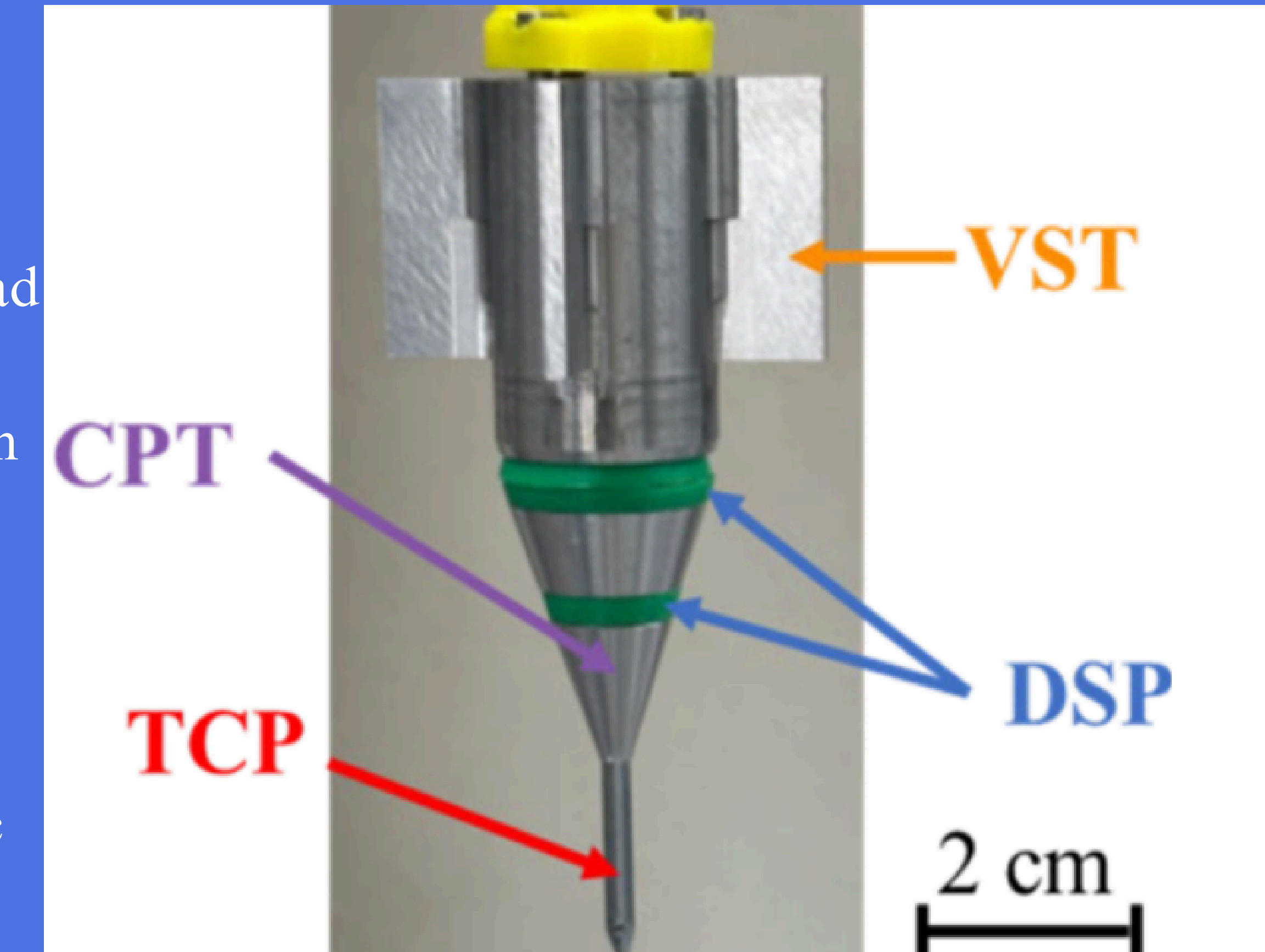


Figure 2. SPARTA Probe

IMPROVEMENTS MADE TO THE SYSTEM

MECHANICAL CONTRIBUTION



Figure 3. Stabilizing Rail

- Developed stabilizing rails for the test bench to guide the drill and limit any drift. Custom Parts are modeled in Solidworks and 3D printed to test and will be machined later. This component was created to fix the testbench issue with swaying when the drill is in the air and causing data inaccuracies. As shown in figure 3.
- This was done in conjunction to the software improvement made in figure 4 to further improve visibility and readability of the graphs given by the testbench

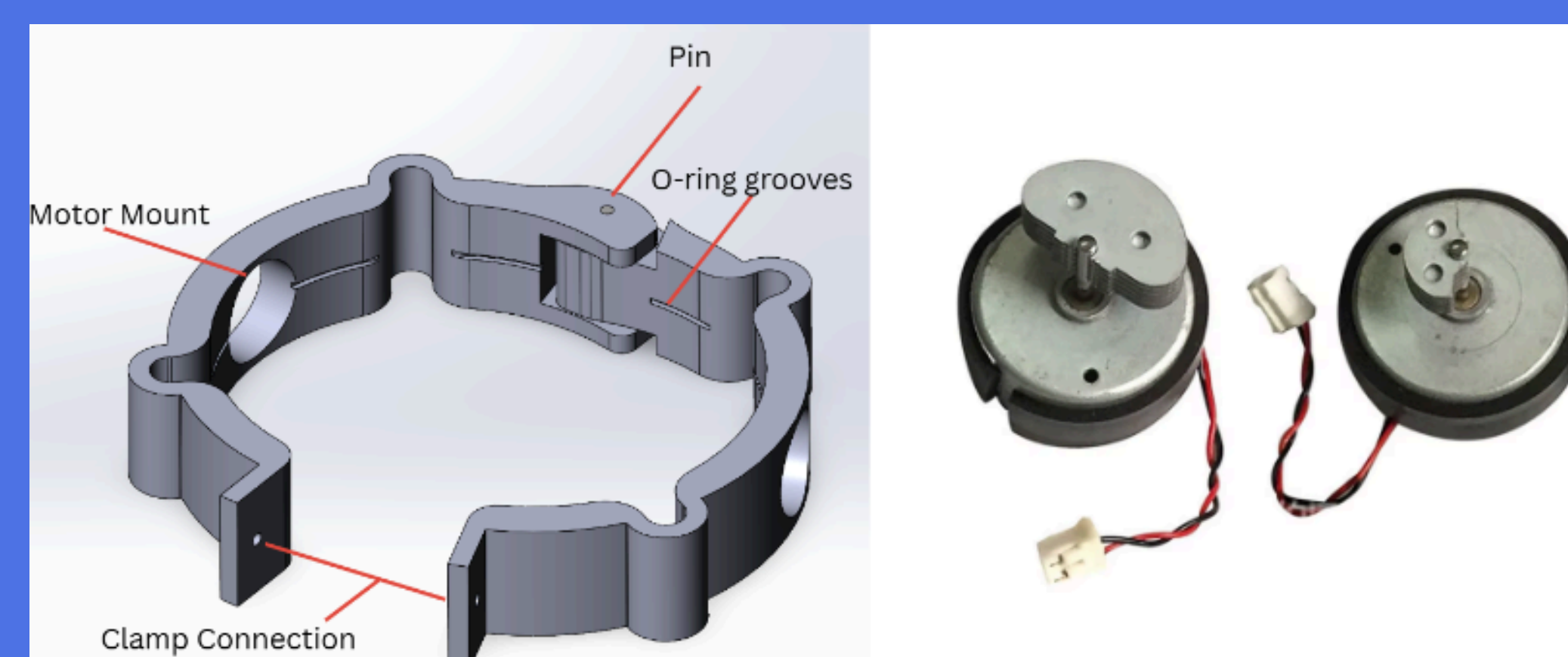


Figure 4. Custom tailored components for Blue Origin SPARTA Payload

- Custom clamps made to hold small rumble motors rated for consistent vibration. O-ring grooves are to provide a tight seal connection to the container so it doesn't move when vibrated. Open the motor mount to give a good contact between the motor and container. Clearance grooves are made to avoid interference with a metal rod pillar inside the system.

SOFTWARE IMPROVEMENTS

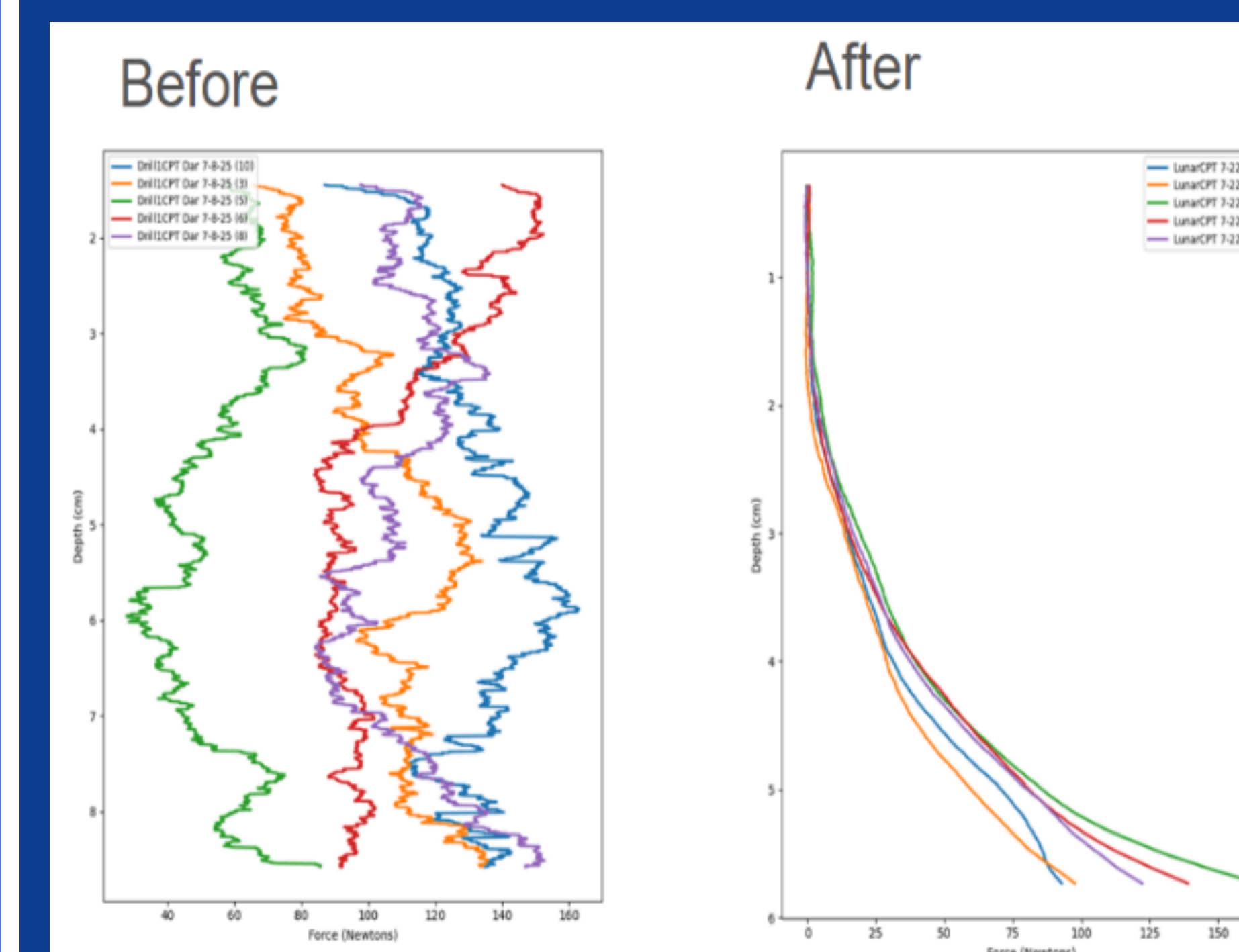


Figure 5. Software Improvements

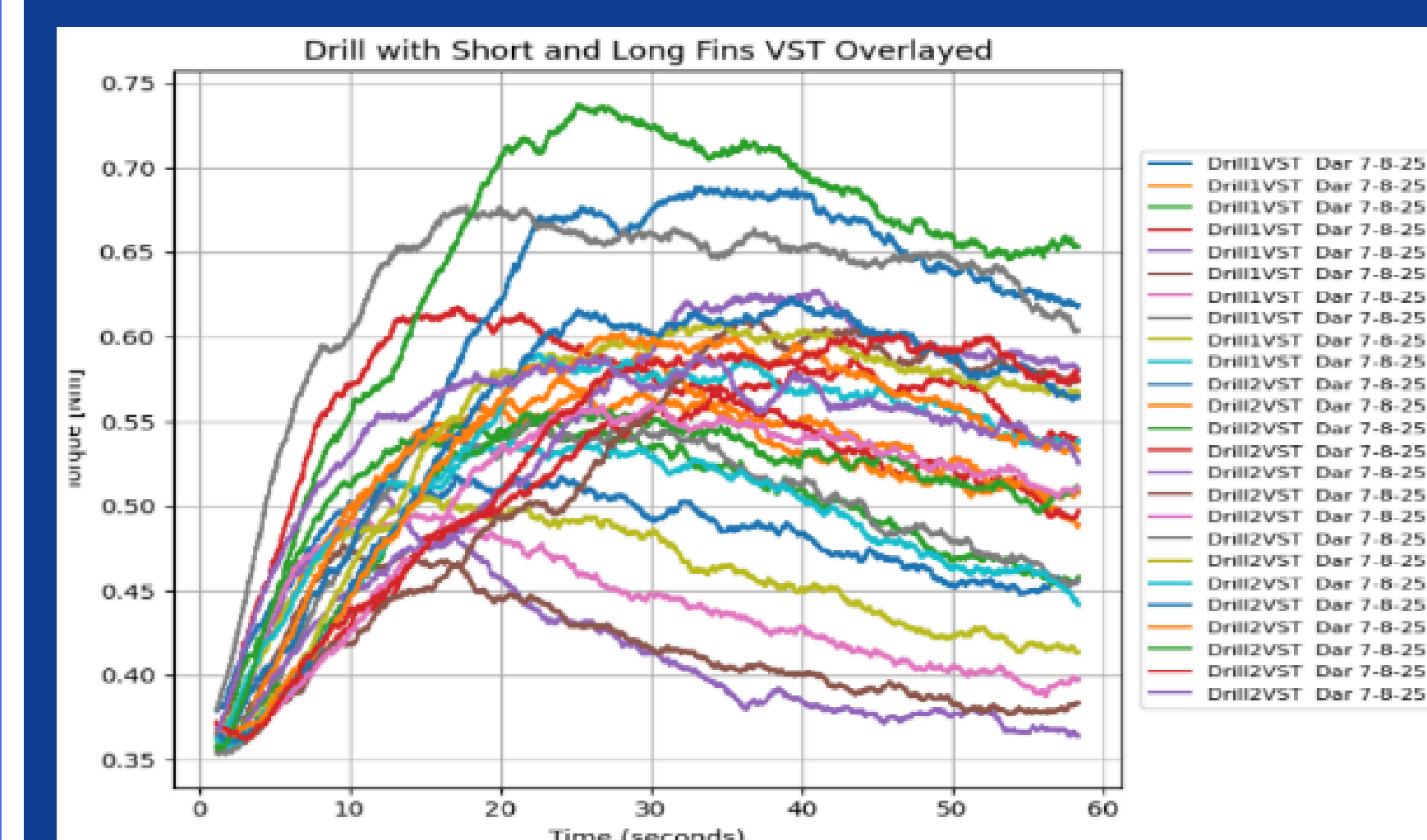


Figure 6. File Parser

- Several software updates were made to improve sensor accuracy and reliability across load, torque, and thermal measurements. Using NI cDAQ and Python, the data pipeline was refined to reduce noise and improve consistency. Key additions include automatic zero/tare routines before each test run, enhanced signal filtering, and refined numerical calculations for improved repeatability and data accuracy.

Created a script to parse several csv files from test run to overlay them on top of each other. Two CSV File Parser that can parse and overlay 100 files every 30 min. This file parser helps see the difference between each test run a lot easier and will be integrated with the software to make it Autonomous.

ABSTRACT

- Hybrid gas-particle combustion is important for metallized propulsion and dust-explosion safety, but there are few small-scale burners that can produce well-characterized, quasi-steady dust clouds for detailed study.
- This work presents a quasi-steady powder-burner test bench that injects controlled solid particles into a premixed methane-air stream while maintaining optical access at the nozzle for flame imaging/diagnostics.
- The design matured through four iterations emphasizing leakage control, manufacturability, sample-loading clearance, and reliable actuation, culminating in a buildable lower-section architecture with defined interfaces for future modules.

MOTIVATION

- Hybrid gas-particle combustion is important for metallized propulsion and dust-explosion safety, but small-scale burners that generate well-characterized, quasi-steady dust clouds are limited.
- This thesis designs a quasi-steady powder-burner test bench that meters particles into a premixed methane-air stream while preserving optical access at the nozzle.
- The design advanced through four iterations focused on sealing, manufacturability, loading clearance, and actuation, producing a complete lower-section layout with defined interfaces for future modules.

DESIGN OBJECTIVES

- Create a quasi-steady powder burner that injects particles into premixed methane-air while maintaining nozzle optical access.
- Achieve precise, repeatable powder feed control with leak-tight sealing and manufacturable/serviceable interfaces.
- Improve safe sample loading and reliable actuation (torque-sized using O-ring contact/friction estimates) and define interfaces for future upper-section development.

ACKNOWLEDGEMENTS

- Advisor: Professor Joseph Kalman
- MAE Department, CSULB

TESTBENCH OVERVIEW

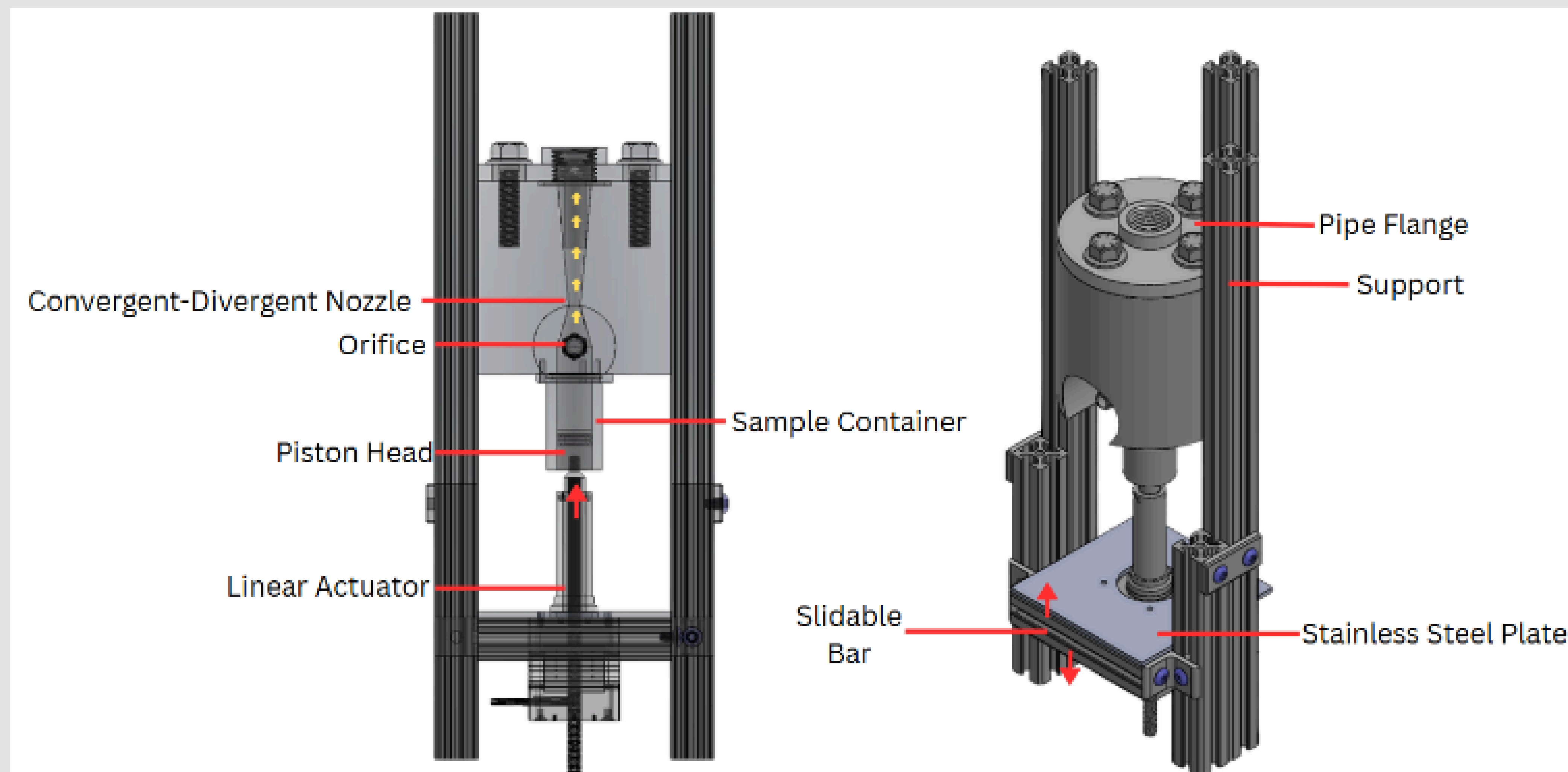


Figure 1. Particle Disperser Testbench

- The testbench is a quasi-steady powder burner built around a rigid vertical frame supporting the flow hardware and sample feed mechanism.
- Methane enters through a flanged inlet, metered through a small orifice into a converging-diverging Venturi section that acts as an ejector mixer, creating low pressure to promote particle entrainment and mixing.
- A sealed sample container is driven upward by a variable-speed linear actuator and piston to deliver controlled powder into the throat region, with the lower assembly mounted on a sliding platform for repositioning during loading and maintenance.

FEATURES

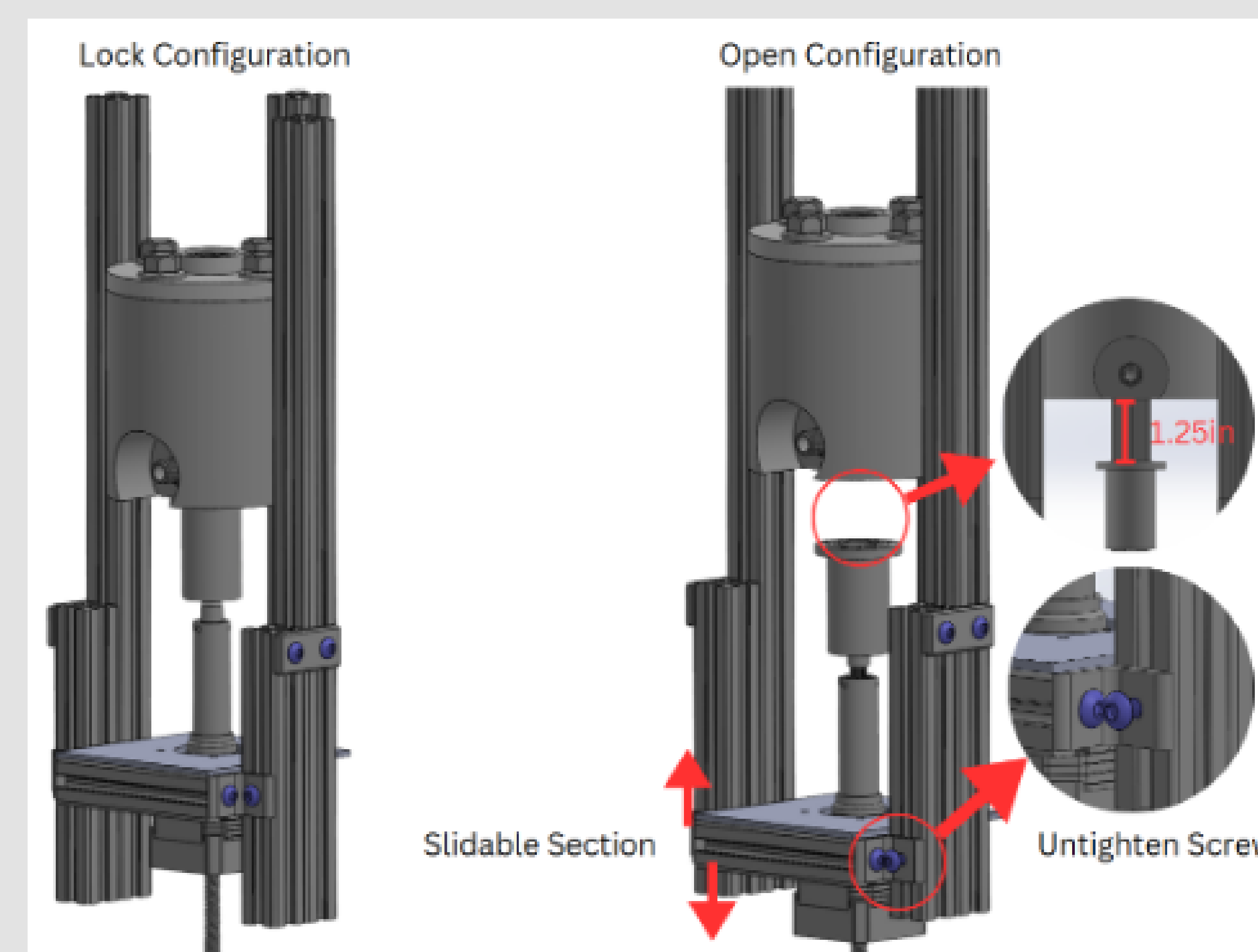


Figure 2. Slidable Platform

The testbench uses a slidable lower platform to make loading and maintenance faster and safer. By loosening the fastening screws, the bottom section can be shifted downward to create a larger access gap for inserting the sample container and handling powdered mixtures. Once the sample is loaded, the platform is raised back into position and locked to restore a rigid, well aligned structure for repeatable operation.

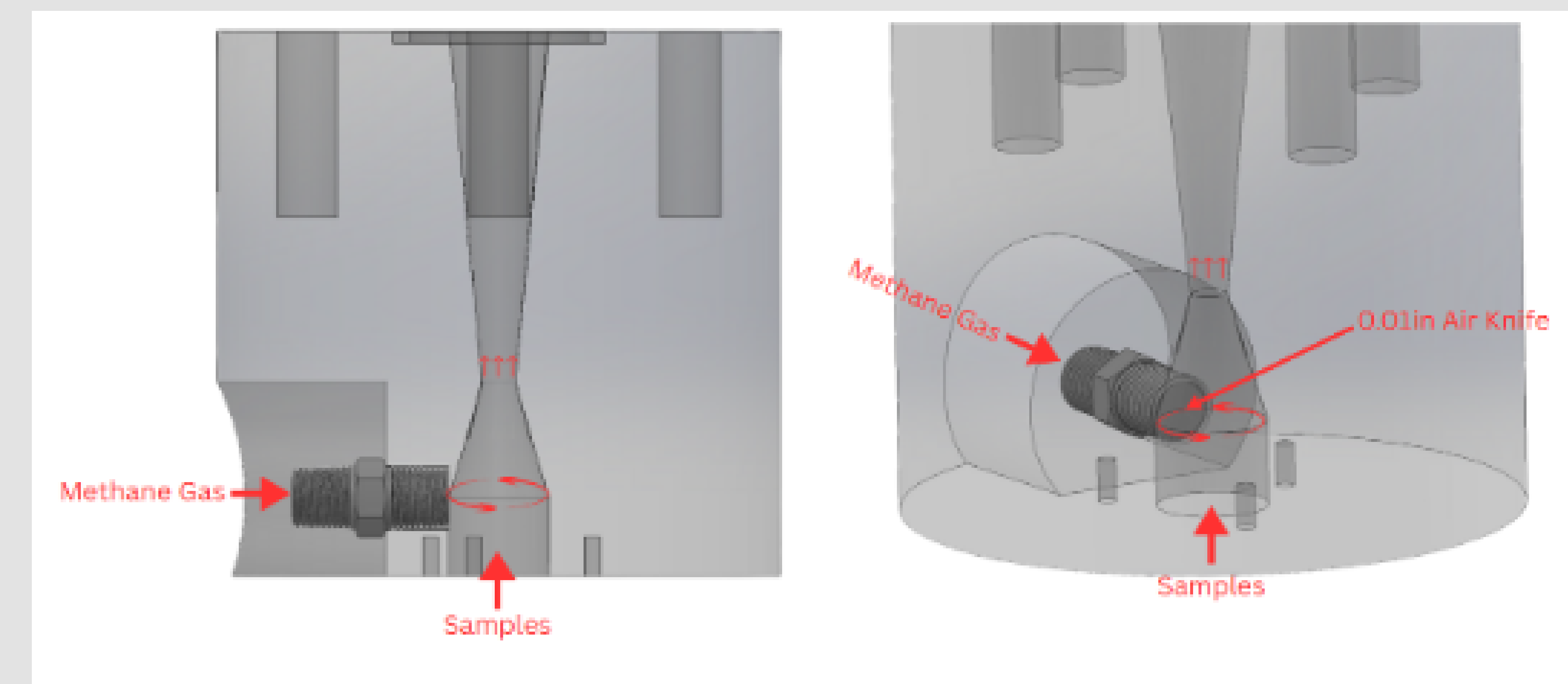


Figure 3. Airknife

Methane is introduced through a small orifice to form a thin, high velocity airknife that sweeps across the Venturi throat. This jet helps create a strong local shear layer that pulls particles into the low pressure region, breaks up clumps, and promotes consistent entrainment of the powder into the main flow. The result is more uniform mixing of the powdered sample with the methane air stream before the mixture expands through the diverging section.

ABSTRACT

- The FireEye UAV is a hybrid VTOL (Vertical Takeoff and Landing) fixed-wing aircraft designed to support wildfire detection, monitoring, and reconnaissance. The system combines quad electric VTOL propulsion for takeoff, landing, and hovering with a gasoline-powered pusher engine for long-endurance cruising.
- Using MATLAB-based propulsion analysis tools and CAD-derived mass properties, the team developed an aircraft capable of sustaining 4+ hours of flight, carrying a 10-kg payload, all while meeting the 100-kg MTOW requirement.
- This hybrid architecture enables precise operations around active wildfire regions, rapid deployment, and reliable loiter capability for continuous surveillance awareness.

MOTIVATION

- More frequent, intense wildfires demand rapid detection and continuous monitoring that current manned aircraft and UAVs cannot reliably provide, motivating the FireEye UAV, which combines VTOL flexibility with fixed-wing efficiency to reach remote fire zones without runways and sustain long-endurance environmental monitoring.

DESIGN OBJECTIVES

- Develop a hybrid UAV capable of vertical takeoff/landing, efficient cruise, and robust endurance.
- Achieve ≥ 4 hours of cruise time with 15% fuel reserve.
- Support 100 kg MTOW including airframe, battery, fuel, avionics, and payload.
- Maintain cruise speed of 25 m/s for optimal coverage.
- Integrate a 10 kg payload bay suitable for a fire suppressant payload.
- Optimize propulsion for minimum subsystem mass while satisfying thrust, climb, and reserve requirements.

ACKNOWLEDGEMENTS

Our Team Phoenix 1 thanks:

- Advisor: Professor Praveen Shankar
- MAE Department, CSULB
- Our instructors and peers for continuous feedback throughout MAE 478 & 479

DESIGN METODOLOGY AND COMPONENTS DESIGN

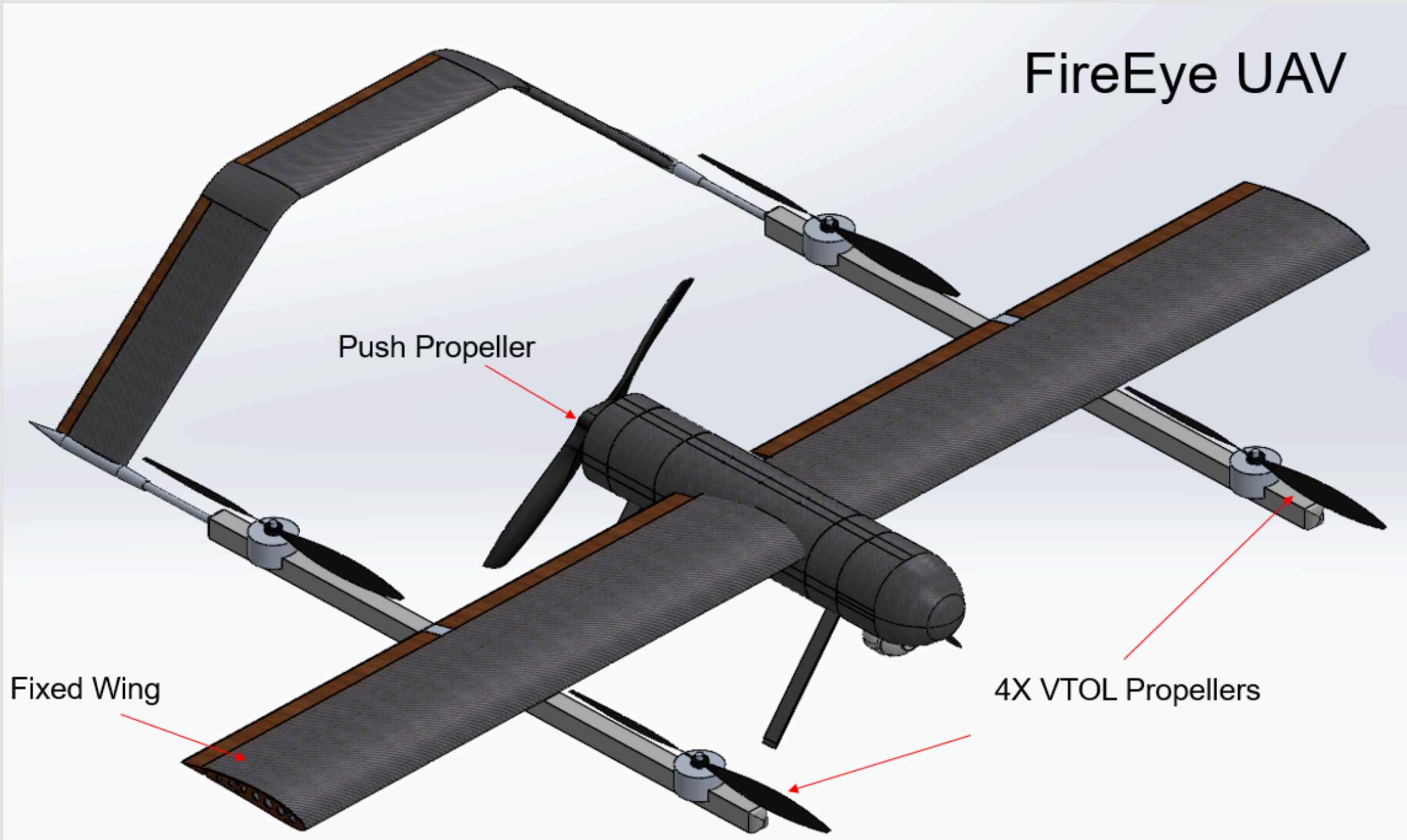


Figure 1. Fire Eye UAV Assembly

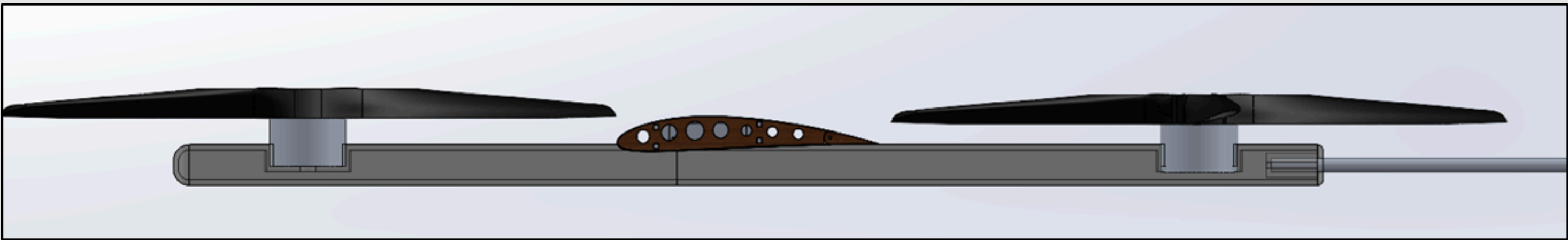


Figure 2. VTOL Boom Design

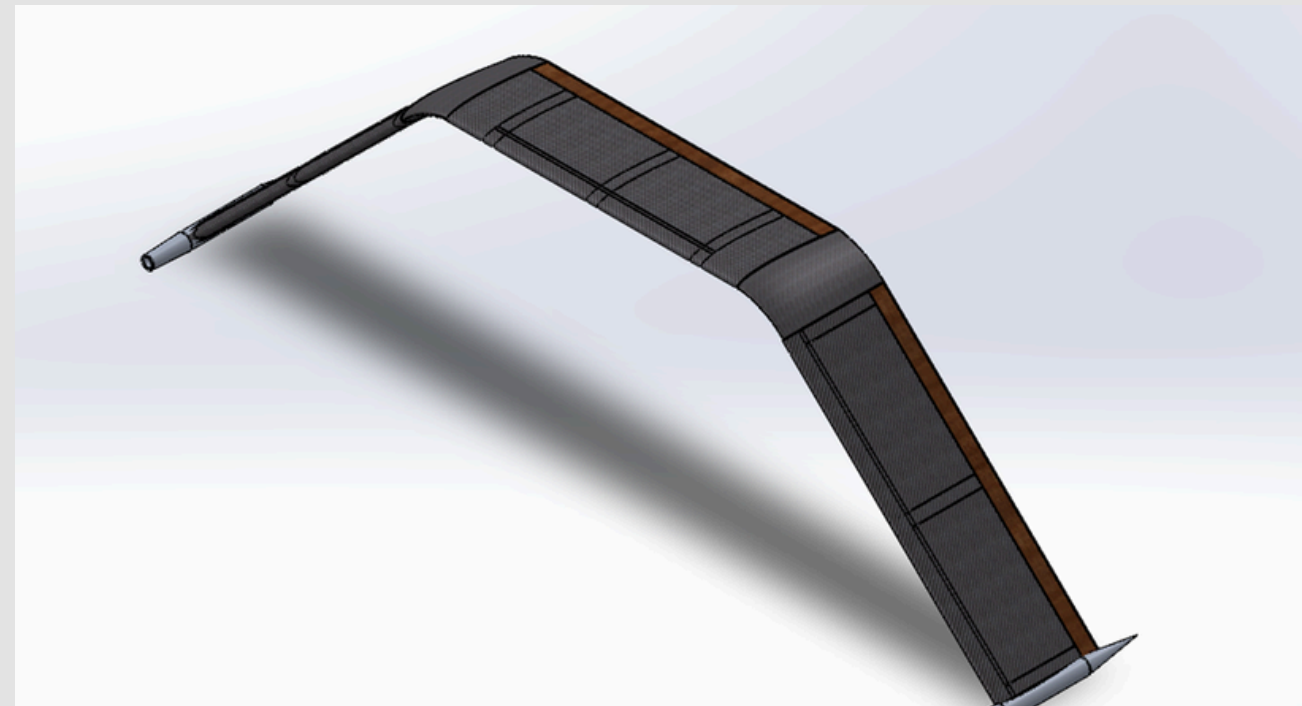


Figure 3. V Tail Design

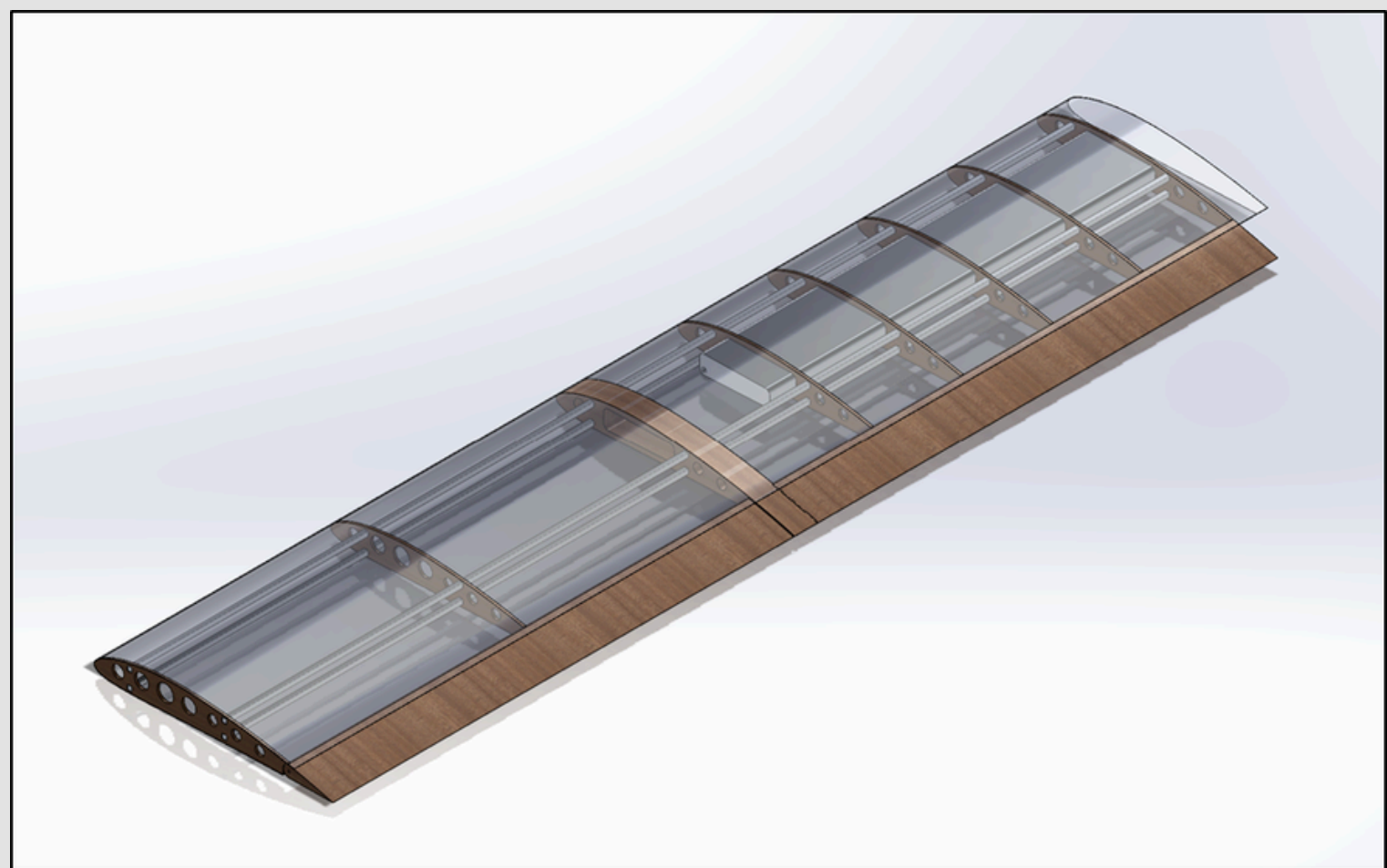


Figure 4. Wing Construction

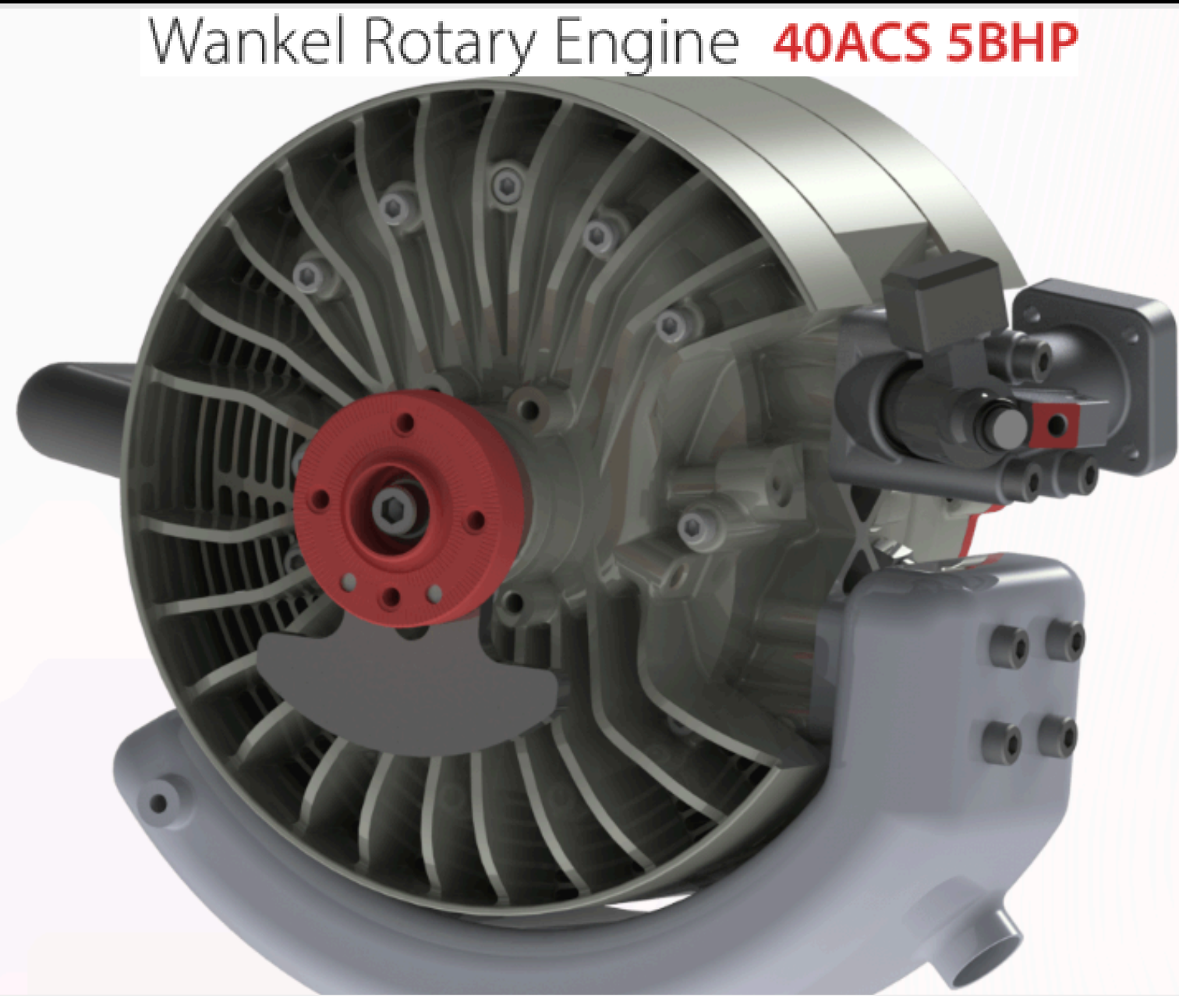
- The design process began by defining mission requirements and analyzing similar UAV systems, followed by iterative weight estimation to set initial sizing. Key aerodynamic and performance parameters (wing loading, drag polar, thrust needs, and endurance targets) were calculated to guide configuration selection.
- MATLAB-based VTOL and cruise propulsion models were integrated with CAD mass estimates to refine MTOW, thrust, power, and fuel/battery requirements. The system was iterated until all mission requirements (VTOL lift and climb, coverage area of 25 sqkm, and ≥ 4 -hour endurance) were satisfied with proper margins.

Parameter	Value
Wing Area	2.5 m ²
Wing span	4.3 m
Aspect Ratio	10
Total length	2.98 m
Payload Capacity	10 kg
Coverage Area	25 sq. km
MTOW	100 KG
Calculated TOW	93 KG

PROPULSION SIZING



Parameter	Value
Prop diameter	VZ 42*16.5"
Req. thrust (climb)	31.16 kgf
Per-rotor thrust (design)	35.12 kgf @ 60% throttle
Climb time to 122 m	48.6 sec
Battery energy (24S)	7.10 kWh
VTOL System	20.75 kg



Parameter	Value
Prop diameter	Mejzlik 32x12
L/D	11.2
P_req (15% power margin)	3.45 kW
Engine avail	3.66 kW
Fuel mass (4 h + 15%)	5.56 kg (7.52 L)
Cruise system (Gasoline)	8.90 kg

AVIONICS SUBSYSTEM SELECTION

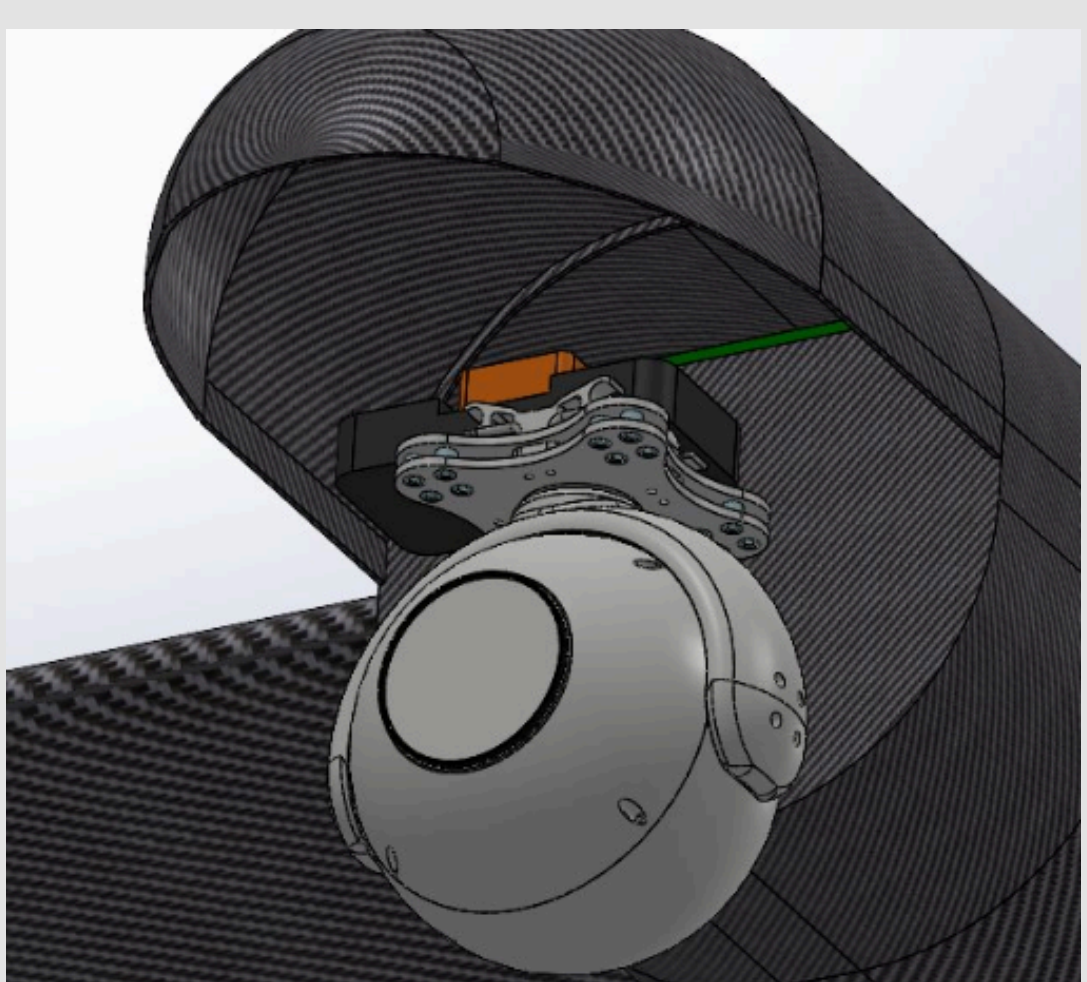
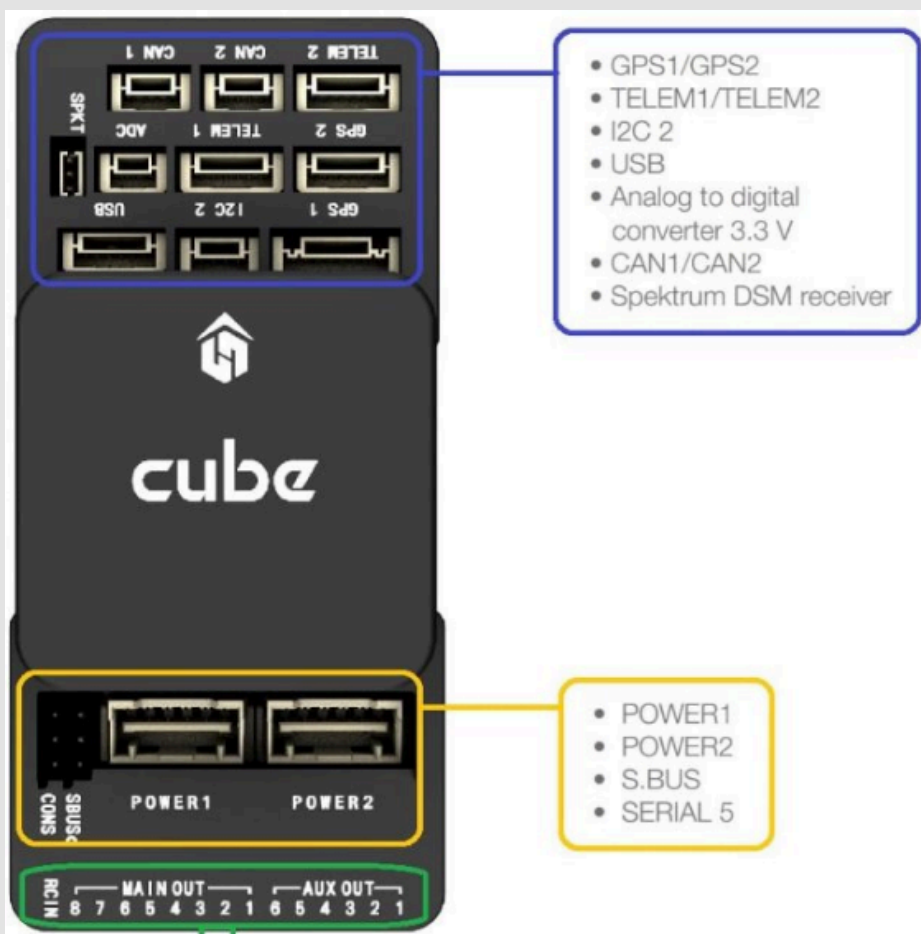


Figure 5. Avionics Bay Bottom View

Figure 6. Avionics Board



PAYLOAD SUBSECTION DESIGN

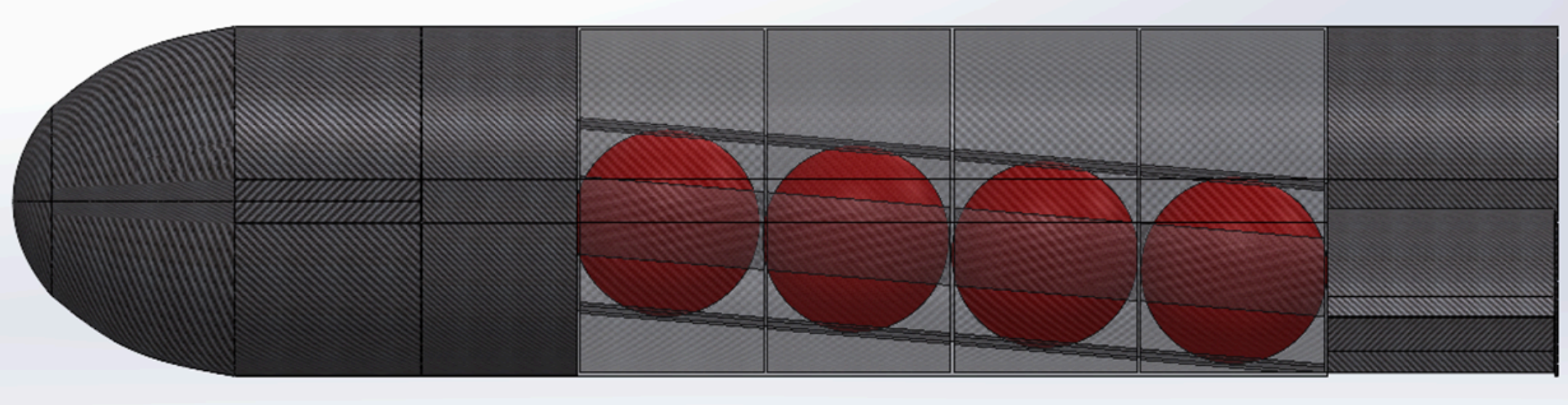


Figure 7. Ramped Deployment Mechanism

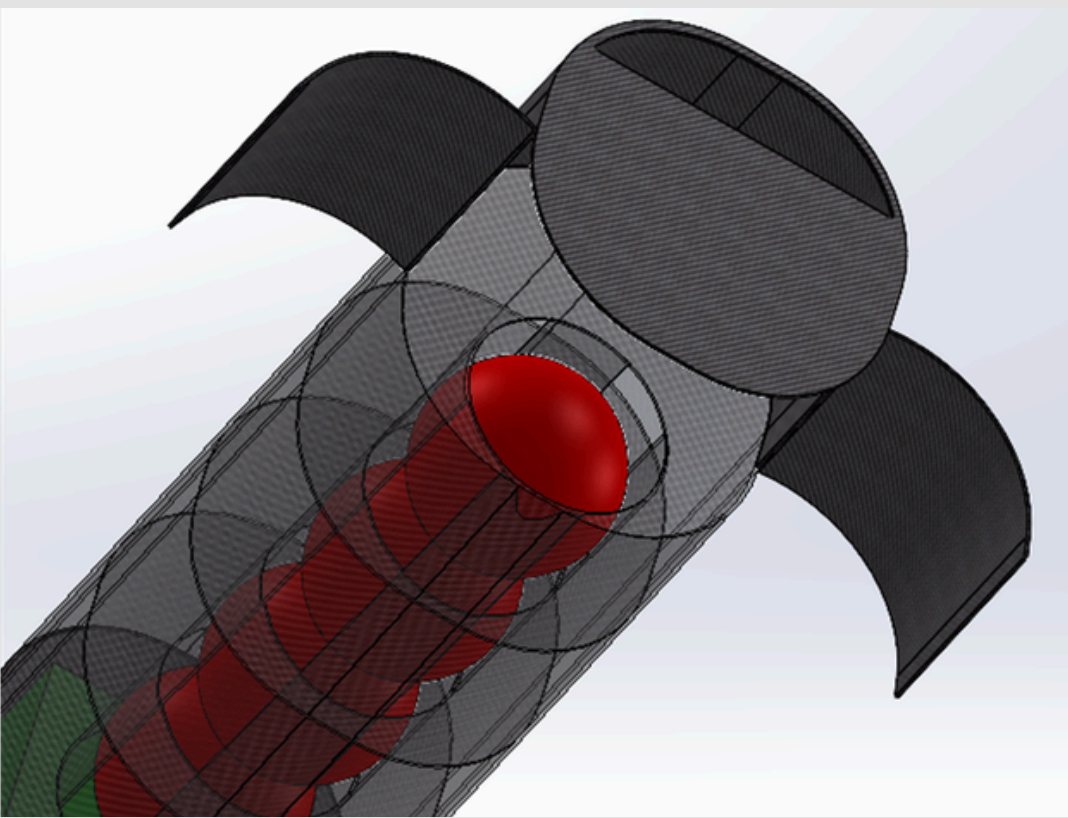


Figure 8. Rear Door Exits

EXPECTED PERFORMANCE

Parameter	Value
Cruise Speed	25 m/s
Endurance	4 hours
Cruise Altitude	122 m
Turning Radius	2.59 m
Stall Speed	18.2 m/s
Dash Speed	25.09 m/s
Maximum Forward Thrust (at h=122m)	117.65 N
Maximum Vertical Thrust (Sea Level)	1823.662 N = 185.9 kgf

ABSTRACT

- This work presents the preliminary aerodynamic design of a subsonic wind tunnel intended for experiments in lift, drag, and basic external aerodynamics.
- The tunnel features a $1\text{ m} \times 1\text{ m}$ test section operating at a design Mach number of 0.6, sized using quasi-one-dimensional isentropic analysis and established design guidelines.
- The final geometry satisfies laboratory space constraints and is validated using 2-D CFD simulations, showing good agreement with analytical predictions.

MOTIVATION

- Wind tunnels provide repeatable, real-world aerodynamic data that complements and validates computational methods.
- Many aerodynamic phenomena such as separation, stall, and wake behavior remain difficult to capture accurately with CFD alone.
- A dedicated subsonic wind tunnel enables controlled testing of airfoils, wings, and bluff bodies under realistic flow conditions for research and education.

DESIGN OBJECTIVES

- Deliver a steady, uniform subsonic flow in a $1\text{ m} \times 1\text{ m}$ test section at Mach 0.6 for external-aerodynamics experiments.
- Size the contraction and diffuser to balance flow quality, pressure recovery, fan power, and overall tunnel length within space constraints.
- Validate the preliminary design using quasi-1D analysis and CFD, establishing a foundation for future structural and detailed design work.

DESIGN METHODOLOGY AND OVERVIEW

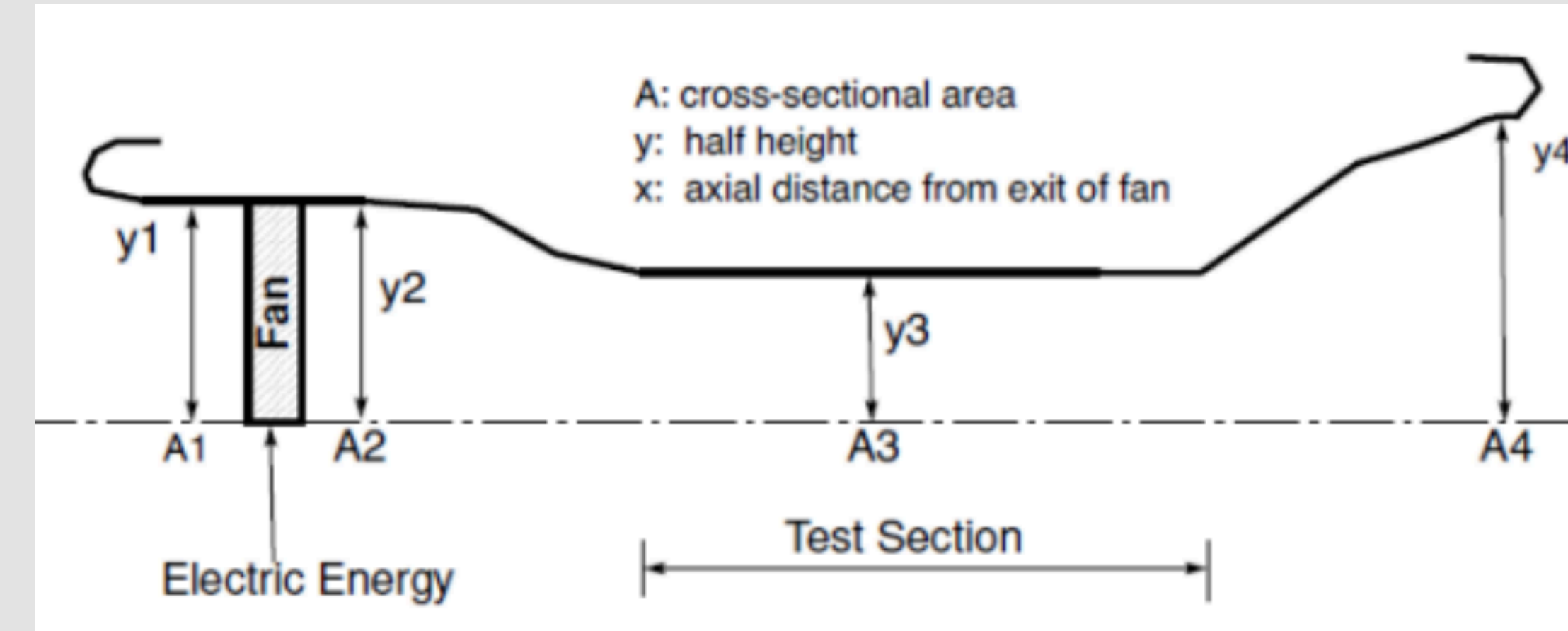


Figure 1. Design Sketch of windtunnel

- The wind tunnel was designed using a quasi-1D approach, linking area changes to target flow conditions and facility constraints. The contraction and diffuser were sized using established aerodynamic guidelines to manage flow quality and pressure recovery. A smooth wall contour was generated in Python via a cubic half-height function and iterated until the layout fit within the lab envelope. The final geometry was then validated with 2D CFD to verify the velocity field and check for separation or nonuniformity.

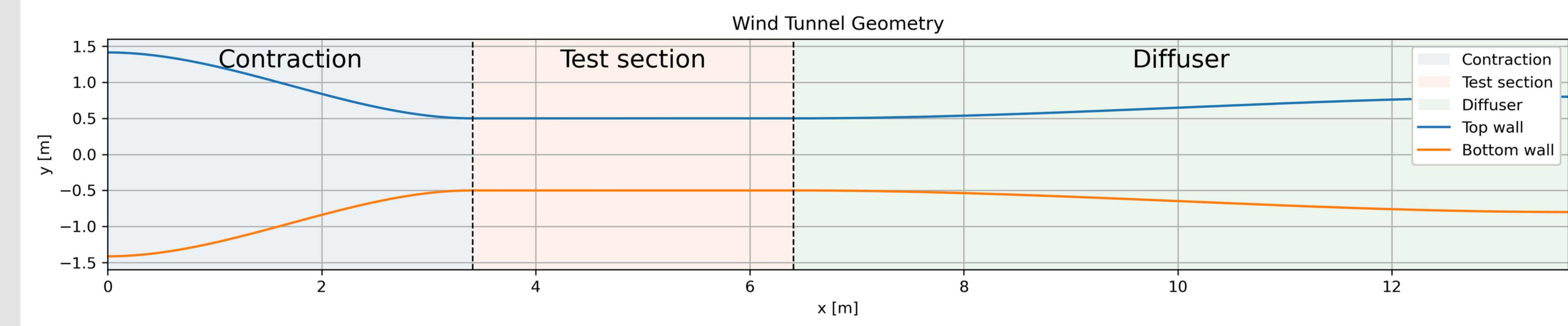


Figure 2. Python generated wind tunnel

- Figure 1 shows the overall tunnel layout from the fan through the contraction, test section, and diffuser. Figure 2 shows the Python-generated contour, highlighting the contraction into the constant-area test section and the subsequent diffuser expansion. Together, they illustrate a complete flow path designed for uniform test section flow with manageable pressure losses and a compact footprint.

ASSESSMENT AND VERIFICATION

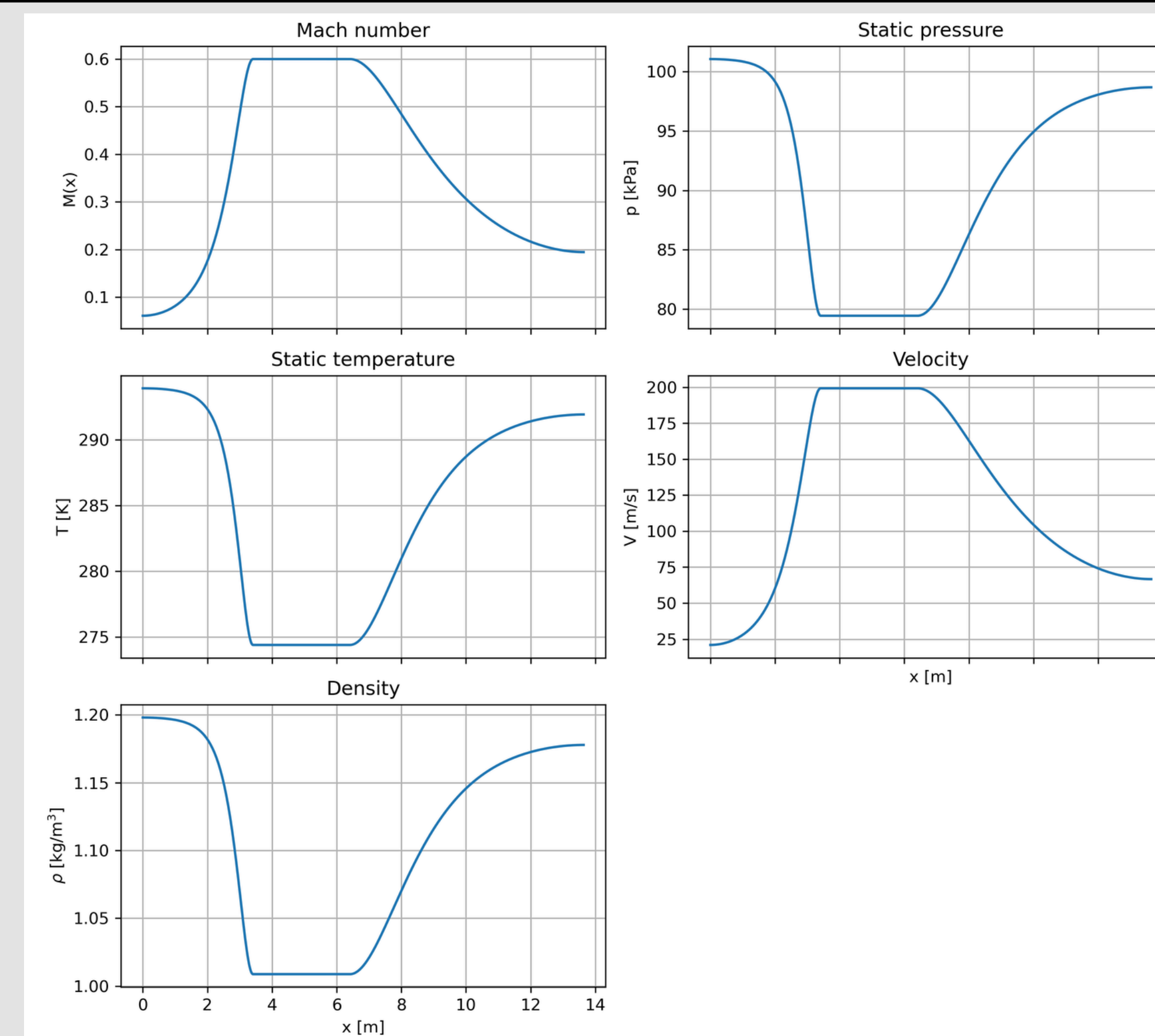


Figure 3. Python Assessment

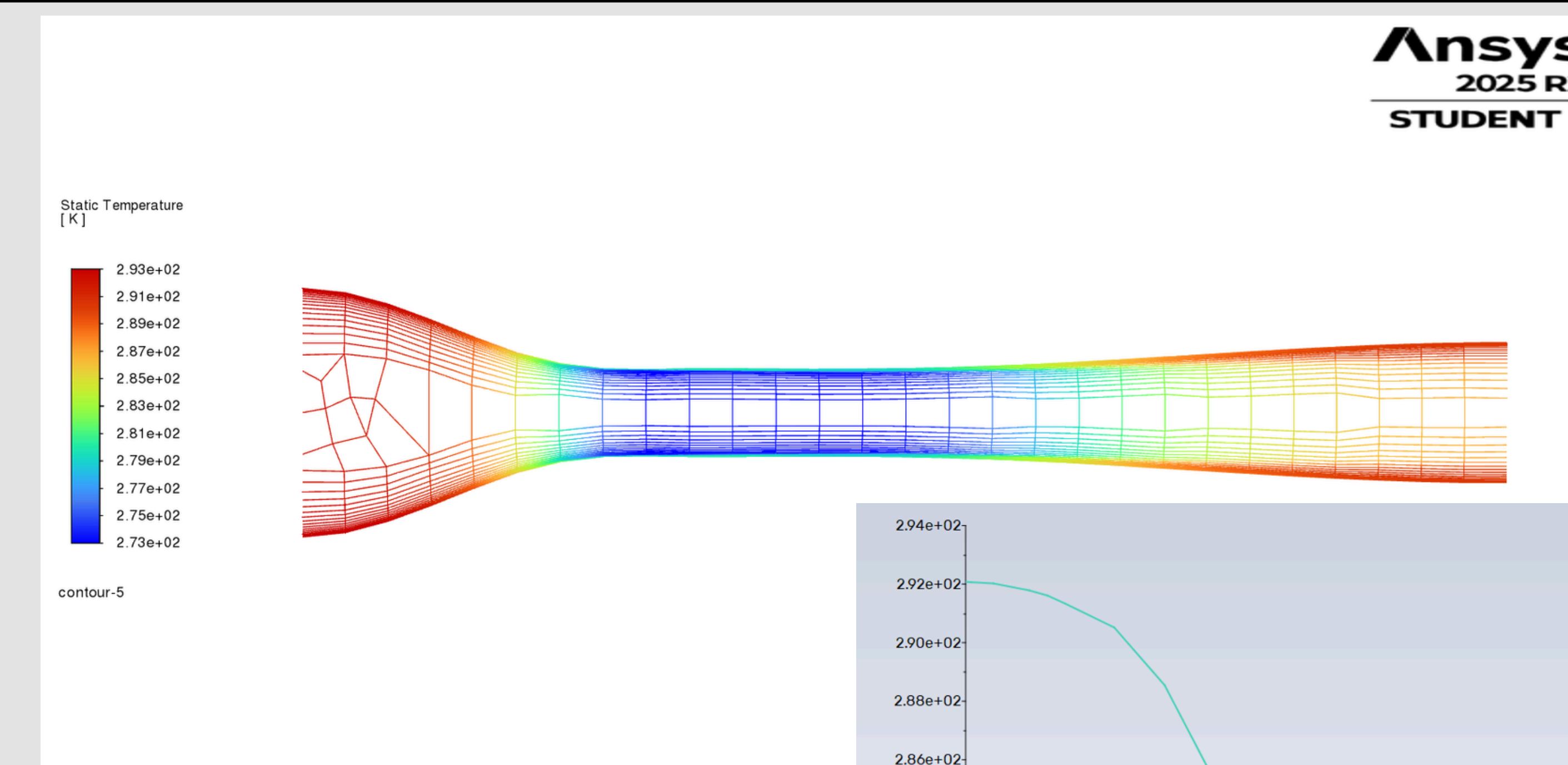
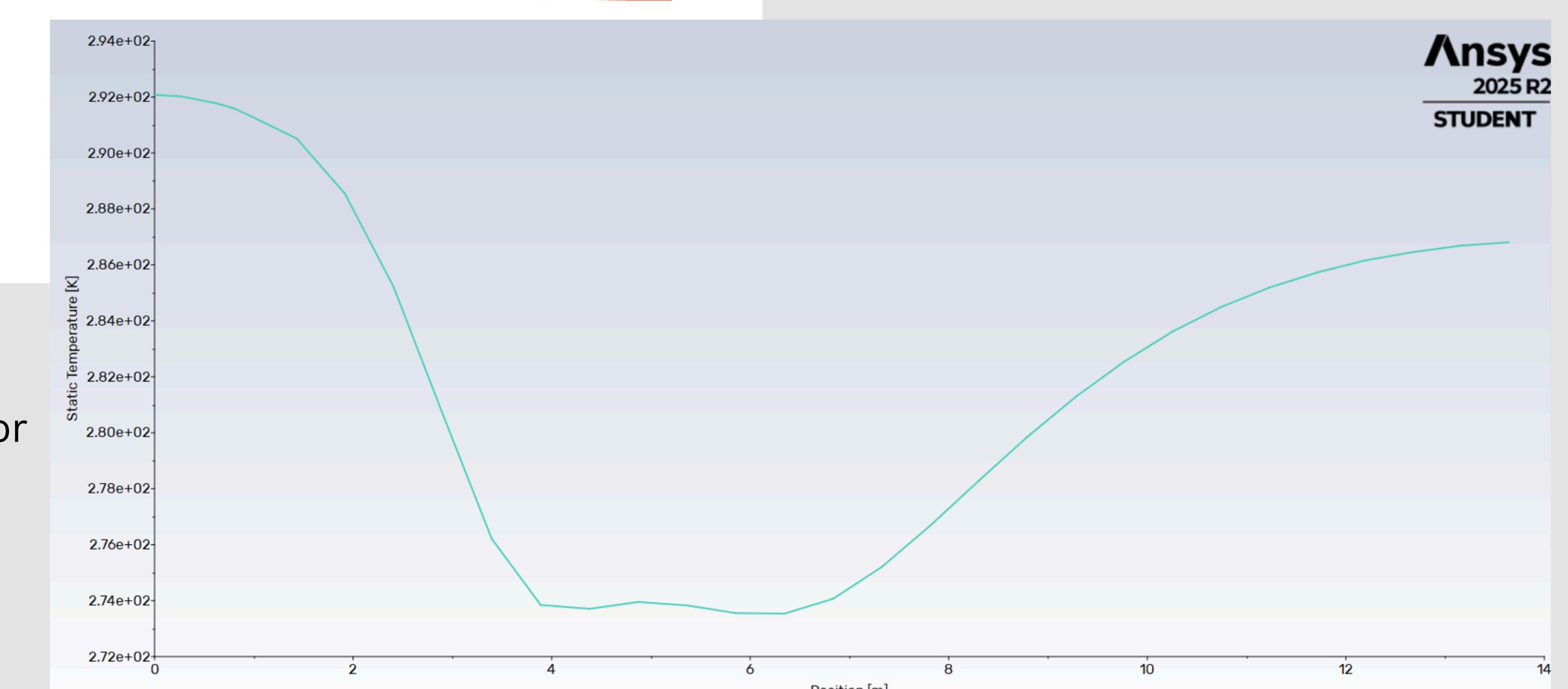


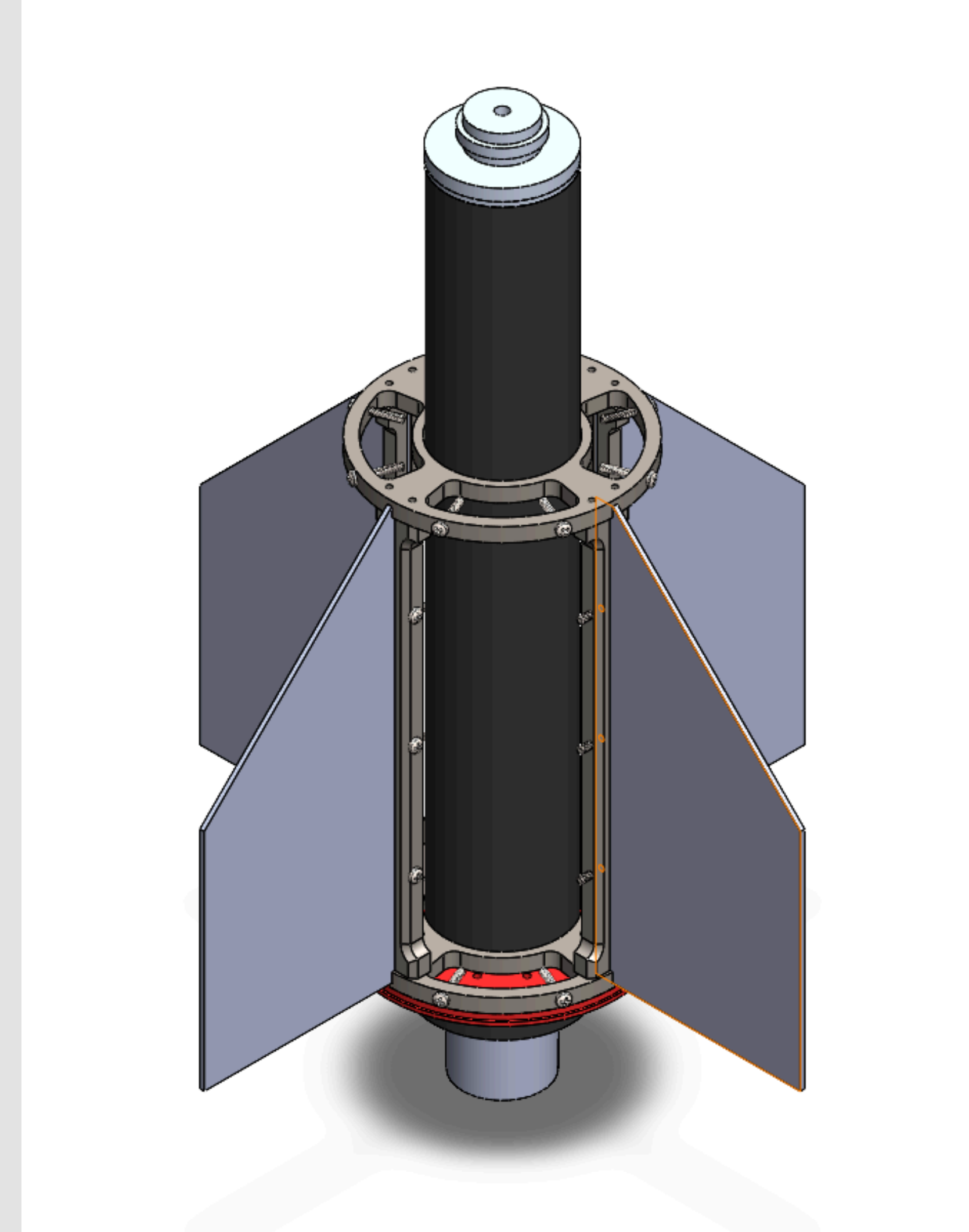
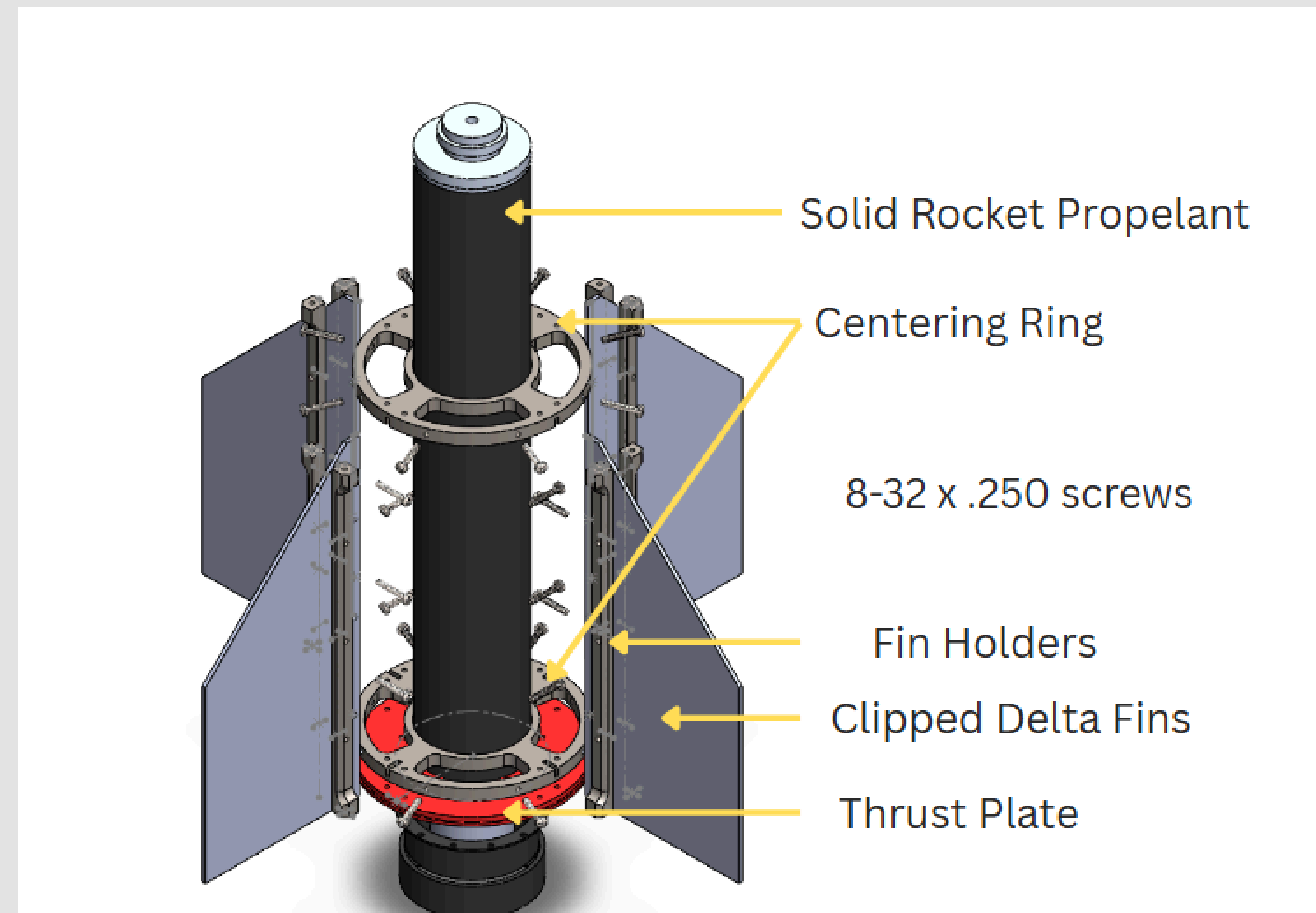
Figure 4. ANSYS contour plot for static temperature

Figure 5. ANSYS centerline plot for static temperature



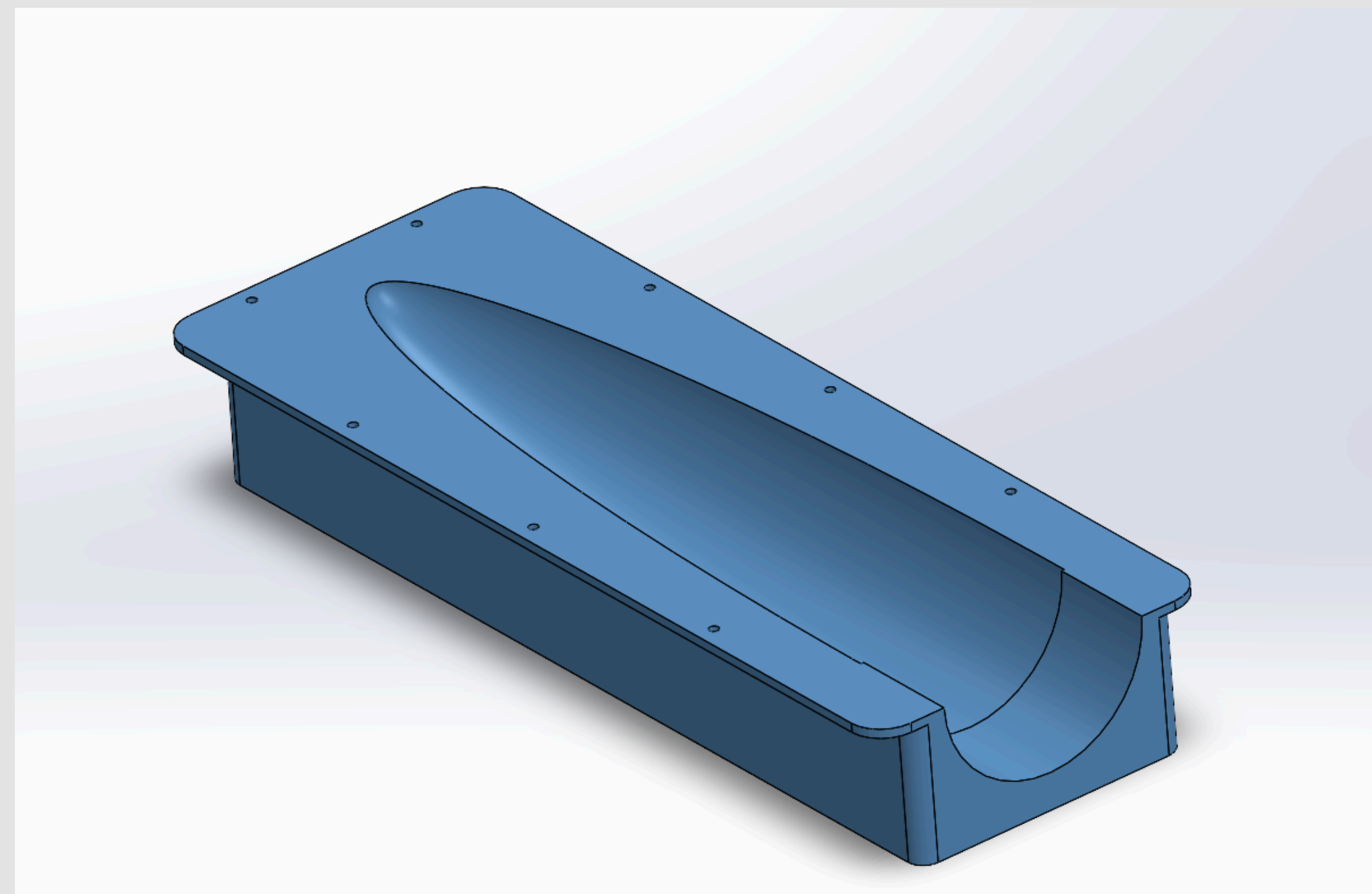
- Figures 3 to 5 present the assessment and verification of the wind tunnel flow behavior using both a Python based quasi one dimensional model and Ansys CFD. In Figure 3, the Python assessment predicts a smooth acceleration through the contraction to the target test section condition, followed by nearly steady properties through the constant area test section and a gradual deceleration with pressure and temperature recovery in the diffuser. Verification was then performed in Ansys by examining both the static temperature contour (Figure 4) and the centerline static temperature profile (Figure 5). The CFD results reproduce the same overall trends and transition regions seen in the Python output, providing confidence that the tunnel geometry and flow assumptions produce the intended test section environment and that the diffuser expansion recovers conditions without introducing major nonuniform behavior.

Reusable Fin System



- Previous designs had fins bonded directly to the airframe, which proved problematic as repeated launch loads caused fin damage and since the fins and airframe were integrated, replacing damaged fins required replacing the entire airframe assembly.
- This modular fin retention system was developed to address that, allowing fins to be individually removed and replaced without disturbing the airframe, reducing both repair time and cost.

Manufacturing and CAD of A Nose Cone



- As part of an experimental fabrication effort, the team attempted to manufacture a fiberglass nose cone using a hand layup process. A SolidWorks model was created to define the nose cone geometry, which was then used to guide the mold and layup process.
- The team applied fiberglass sheets and resin by hand, iterating on technique to achieve the desired shape and structural integrity. While experimental, this process provided valuable hands-on experience with composite manufacturing methods.

